STAT0023 ICA2

Group 26

1. Introduction

The aim of this project is to investigate the variation in COVID-19 death counts across MSOAs in England and Wales between March and July 2020, and to develop a model for predicting missing death counts.

Official statistics from the Office for National Statistics[1] show clear spatial and socioeconomic inequalities: COVID-19 mortality in the most deprived areas was more than twice that in the least deprived areas, while London and other urban areas had substantially higher mortality than rural regions. Individual-level evidence also highlights the importance of demographic and health-related risk factors, including age, ethnicity and comorbidities[2]. These patterns motivate the use of demographic, socioeconomic and spatial predictors in a count-based modelling framework.

Motivated by these patterns, this report conducts an exploratory analysis to identify key demographic, socioeconomic, and spatial factors associated with COVID-19 mortality at the MSOA level. These insights are then used to guide the specification of an appropriate count-based model for prediction.

2. Data overview

The dataset UKCovidWave1.csv contains 7,201 MSOAs in England and Wales, of which 5,401 have observed COVID-19 death counts and 1,800 have missing values (Deaths $= -1$) to be predicted.

A wide range of variables is provided, covering geographical structure (Region, RUCode), population characteristics (e.g. density, age structure), and socioeconomic factors such as deprivation, health status, and ethnicity. These variables are well motivated by existing evidence showing that COVID-19 mortality is driven by a combination of demographic vulnerability, socioeconomic inequality and spatial factors.

To facilitate modelling, a smaller set of derived variables is constructed to summarise these factors, allowing the analysis to focus on key drivers while reducing unnecessary complexity.

3. Data preprocessing

Derived variables were constructed to summarise key demographic, socioeconomic and spatial mechanisms while reducing the dimensionality of the original dataset. Since MSOAs differ substantially in population size, most count variables were converted into proportions to allow fairer comparison across areas. We omit some intermediate variables and only present the variables that might be used in the modelling process. The variable log_pop was also created from total population and later used as an offset in count models.

The derived variables capture several mechanisms supported by existing evidence: age structure reflects the higher vulnerability of older populations; care-home and communal-residence variables capture exposure in high-risk settings; deprivation, household health problems and education-related variables represent

[1] ONS Statistical Bulletin *Deaths involving COVID-19 by local area and socioeconomic deprivation: deaths occurring between 1 March and 31 July 2020* (August 2020).

[2] Williamson et al. (2020): *Factors associated with COVID-19-related death using OpenSAFELY* (Nature 584, pp. 430–436).

socioeconomic vulnerability; transport and density variables proxy mobility and urban exposure; and ethnicity/migration indicators reflect population heterogeneity associated with COVID-19 outcomes. These choices are consistent with evidence that COVID-19 mortality varies by age, deprivation, ethnicity and geographical context, and with ONS findings on strong spatial, urban–rural and deprivation-related mortality differences[3].

Overall, these derived variables provide a more interpretable starting set for exploratory analysis and subsequent modelling, while avoiding excessive complexity from highly disaggregated original variables.

4. Exploratory data analysis

The purpose of this exploratory analysis is to identify a sensible set of candidate variables for later modelling, and to understand whether the data structure suggests a linear model, a count model, or a more flexible approach.

4.1 Distribution of the response

We begin by examining the distribution of observed COVID-19 death counts.

The observed death counts are strongly right-skewed based on the histogram and boxplot. Most MSOAs have fewer than 10 deaths, while a few exceed 30, with a maximum above 60. The mean is 7.22 and the variance is 31.35, indicating potential overdispersion.

This pattern is expected for small-area COVID-19 mortality data, where ONS notes that local death counts can be low and unevenly distributed, making area-level rates unstable[4]. Therefore, ordinary linear regression is inappropriate, and count models such as Poisson or negative binomial regression are more suitable.

4.2 Death counts and population size

Since larger MSOAs may record more deaths simply because they contain more people, raw count variation must be separated from underlying mortality risk, and we use plots and correlation to analyse them.

Deaths_obs is positively related to PopTot, with a correlation of about 0.24, indicating that larger MSOAs tend to record more deaths partly because they contain more people. However, the relationship is only moderate.

After scaling by population, death_rate_per_1000 has almost no relationship with PopTot, with correlation around -0.02. This suggests that population size mainly affects the scale of death counts rather than underlying mortality risk.

Therefore, Deaths_obs is retained as the count response as required, while log_pop = log(PopTot) is used as an offset to account for different MSOA population sizes.

4.3 Geographical variation

Now, we investigate geographical differences.

Death rates vary clearly across both Region and RUCode by boxplots, with differences in medians and spreads suggesting that geographical variation is not driven only by population size or a few extreme observations.

[3]Williamson et al. (2020): *Factors associated with COVID-19-related death using OpenSAFELY* (Nature 584, pp. 430–436).

[4]ONS Statistical Bulletin *Deaths involving COVID-19 by local area and socioeconomic deprivation: deaths occurring between 1 March and 31 July 2020* (August 2020).

This supports the inclusion of spatial predictors in the model. It is also consistent with ONS evidence that COVID-19 mortality differed substantially by region, with London having the highest age-standardised mortality rate, and by urban–rural classification, with urban major conurbations showing the highest mortality[5]. Therefore, Region and RUCode are retained as candidate categorical predictors.

4.4 Candidate variables check

To assess relevance, correlations between candidate variables and the crude death rate are examined to reduce the influence of population size.

Only a subset of variables is retained, guided not only by correlation strength but also by prior evidence and interpretability. In particular, we focus on key mechanisms identified in the literature: age structure (e.g. p_age75), care-related exposure (p_care), socioeconomic vulnerability (p_depr, p_gradeDE), urban exposure (PopDens, p_trans), and population heterogeneity (p_black, p_noneu).

Correlations are generally modest, with the largest for p_care (0.326), followed by p_hlth (0.150) and p_age75 (0.147), while most others are below 0.15. Some variables show weak negative associations (e.g. p_noch: -0.128). Nevertheless, selection is not based solely on these values. Based on Williamson et al.[6], COVID-19 mortality risk is associated with age, comorbidities, deprivation and ethnicity, supporting the inclusion of these variables despite modest univariate correlations.

These variables are therefore retained as candidate predictors for multivariable modelling, where their combined and adjusted effects can be assessed.

4.5 Correlation among candidate predictors

We calculate the correlation matrix between variables and visualise with a heatmap. The heatmap shows moderate correlations among related predictors, especially between age variables, socioeconomic indicators, ethnicity variables, population density and transport use. This suggests some overlap between variables, but no severe pairwise collinearity requiring immediate removal.

These associations also indicate that mortality risk may depend on combinations of factors rather than isolated predictors. This supports later consideration of interaction terms, guided by both the data structure and known links between deprivation, urban exposure and demographic vulnerability in COVID-19 mortality.

4.6 Summary of EDA and initial candidate variables

Overall, the EDA suggests that COVID-19 deaths should be modelled as overdispersed count data, with log_pop used as an offset to account for population size. Spatial variation across Region and RUCode is clear, while age, care-home exposure, deprivation, ethnicity, transport and urban structure are supported as candidate predictors by both the data patterns and existing evidence. These findings motivate a multivariable count model in the next stage.

5. Model building

As suggested by the EDA, we may start by fitting a Poisson GLM model to death, including the offset and the set of candidate variables mentioned in the EDA above.

[5] ONS Statistical Bulletin *Deaths involving COVID-19 by local area and socioeconomic deprivation: deaths occurring between 1 March and 31 July 2020* (August 2020).

[6] Williamson et al. (2020): *Factors associated with COVID-19-related death using OpenSAFELY* (Nature 584, pp. 430–436).

5.1 Poisson GLM

A Poisson GLM is first fitted using all candidate predictors, with log_pop included as an offset to control for population size.

Although the estimated effects are consistent with expectations, strong overdispersion is observed: the deviance and Pearson dispersion statistics are 2.74 and 2.92, both well above 1, and only 26.7% of deviance is explained. This suggests that the Poisson assumption of equal mean and variance is violated, a common issue in COVID-19 mortality data where unobserved heterogeneity and spatial clustering inflate variability.

As a result, the Poisson model is likely to underestimate uncertainty. We therefore retain the same predictors but move to more flexible count models, particularly the negative binomial specification.

5.2 Quasi-Poisson and Negative Binomial GLM

The quasi-Poisson model adjusted for overdispersion but did not improve fit and provides no AIC, so it was not used for final comparison.

A negative binomial model gave a much better fit: residual deviance fell from 14693 to 5793, dispersion was close to 1, and AIC decreased from 33662 to 29730. Given the strong spatial and socioeconomic heterogeneity in COVID-19 mortality reported by ONS[7], the negative binomial model was therefore retained as the baseline for further refinement.

5.2.1 Single-term Selection

Single-term deletion tests were used to assess predictor contribution within the negative binomial model. Most variables increased AIC when removed, especially p_care, p_asian, p_noch, Region and RUCode, supporting their importance.

Some predictors, such as p_hlth and p_trans, showed weaker marginal effects after adjustment, suggesting partial overlap with other variables. However, given prior evidence on health vulnerability, mobility, deprivation and spatial inequality in COVID-19 mortality, these variables were not removed solely on marginal significance .

Overall, the model retained a broad predictor set, with later refinement focusing on reducing categorical complexity and capturing nonlinear or interaction effects.

5.2.2 Hierarchical Clustering

To reduce categorical complexity, hierarchical clustering was applied to RUCode using average covariate profiles. Among the cluster options, the 5-cluster version gave the best fit, slightly improving AIC from 29730.10 to 29725.70 while reducing the residual degrees of freedom cost.

This suggests that some rural–urban categories share similar mortality-related profiles, so RUCluster5 captures the main structure more efficiently than the original RUCode. The same approach was tested for Region, but it did not improve fit; therefore, the original Region variable was retained. Hence, we replace the RUCode with RUCluster5 and call this model NB-GLM.

[7]ONS Statistical Bulletin *Deaths involving COVID-19 by local area and socioeconomic deprivation: deaths occurring between 1 March and 31 July 2020* (August 2020).

5.2.3 Diagnostics

Now, we decide to look at the diagnostic plots of the final negative binomial model NB-GLM.. The plots show that although overdispersion is largely controlled, some structure remains in the residuals, with upper-tail departures in the Q-Q plot and underprediction of larger death counts. This suggests that a linear specification is still too restrictive. Therefore, the next stage considers a GAM, allowing nonlinear effects and possible interactions between key predictors.

5.3 Negative Binomial GAM

5.3.1 Smooth terms

We call the fitted model NB-GAM. Smooth terms were retained only for variables showing clear nonlinearity (edf»1), including p_age75, p_care, p_black, p_noqual, p_noch and p_trans, while approximately linear predictors were kept in parametric form. The model also includes a population offset, Region as a random effect, and clustered RUCode.

The selected smooth terms are all highly significant (edf = 3.7–6.8, p<0.001), confirming that key relationships are nonlinear—particularly for age structure and care home exposure, which are known drivers of COVID-19 mortality. The migration variable p_noneu shows only weak additional contribution (p=0.06), and it is not considered since the first model as it provides relative high p-value, but we reconsider it now because it shows some improvements on other associated variables as well as model statistics like AIC and interpretability.

Model fit improves over earlier specifications, with 29.9% deviance explained and dispersion close to 1 (1.08–1.10), indicating that overdispersion is well controlled. Overall, the NB-GAM provides a more flexible and better-fitting representation of mortality risk, consistent with the multifactor and nonlinear nature of COVID-19 outcomes.

5.3.2 Interaction

Interactions were introduced in the NB-GAM guided by epidemiological evidence that key risk factors act jointly to possible address extreme situations[8]. Two were retained: p_age75 × p_care (significant, p=0.03) and p_depr × p_hlth (marginal, p=0.06), reflecting the combined effects of ageing in care settings and the interaction between deprivation and health. We call the updated model NB-GAM2.

The resulting model (NB-GAM2) improves fit, with 30.5% deviance explained and a reduced AIC = 29465.50, while maintaining dispersion close to 1. Other interactions were not retained due to limited contribution.

Overall, we choose NB-GAM2 as the final model as it provides a parsimonious yet flexible final specification, capturing key nonlinearities and interactions in COVID-19 mortality and forming the basis for prediction.The NB-GAM2 is specified as follows,$s(\cdot)$ denotes smooth terms, $ti(\cdot,\cdot)$ denotes tensor interaction terms, $s(Region, bs = \text{"re"})$ represents the regional random effect, and $\log_pop$ is included as an offset.

$$\begin{aligned} Deaths = {} & p_asian + s(p_age75) + s(p_care) + p_depr + p_hlth + p_comm \\ & + s(p_black) + s(p_noqual) + s(p_noch) + s(p_trans) + p_gradeDE \\ & + s(p_noneu) + s(Region, bs = \text{"re"}) + RUCluster5 \\ & + ti(p_depr, p_hlth) + ti(p_age75, p_care) + \log_pop. \end{aligned}$$

6. Prediction

Finally, we make predictions using NB-GAM2 and output the prediction dataset.

[8]Williamson et al. (2020): *Factors associated with COVID-19-related death using OpenSAFELY* (Nature 584, pp. 430–436).

7. Conclusions

This report developed a count-based modelling framework to explain and predict MSOA-level COVID-19 death counts. The final NB-GAM2 accounts for population size, overdispersion, spatial structure, nonlinear effects and selected interactions, giving the best overall fit with 30.5% deviance explained and AIC around 29465.

The final model suggests that COVID-19 mortality variation is not driven by a single factor, but by a combination of demographic, socioeconomic and spatial risks. Higher proportions of older residents, care-home exposure, black and asian population, low educational attainment, lack of central heating and public transport use showed important nonlinear associations with death counts, and regional/rural–urban structures account for imbalance of death rate. The retained interactions also suggest that risk may be amplified where ageing populations overlap with care-home exposure, and where deprivation coincides with poorer household health.

Overall, the model provides an interpretable basis for predicting missing death counts while highlighting the role of demographic vulnerability, socioeconomic inequality and spatial context in COVID-19 mortality.

8. Limitations

This model has several limitations. Diagnostic plots indicate remaining upper-tail deviation and underprediction of unusually high death counts, suggesting that extreme local outbreaks are not fully captured. Such extremes may arise from clustered infections in care homes or sudden local transmission events, which are difficult to model using area-level covariates alone.

The dataset also omits important drivers of COVID-19 mortality, including healthcare access, testing intensity, policy responses and detailed comorbidities. In addition, ONS highlights that small-area death counts can be unstable and affected by reporting delays, so MSOA-level predictions should be interpreted with caution[9].

Finally, the model captures statistical associations rather than causal effects. Observed relationships may reflect broader structural inequalities rather than direct mechanisms, so the conclusions are reasonable and applicable theoretically while they still needs to be combined with real-life scenarios.

[9] ONS Statistical Bulletin *Deaths involving COVID-19 by local area and socioeconomic deprivation: deaths occurring between 1 March and 31 July 2020* (August 2020).

Graphs and Tables

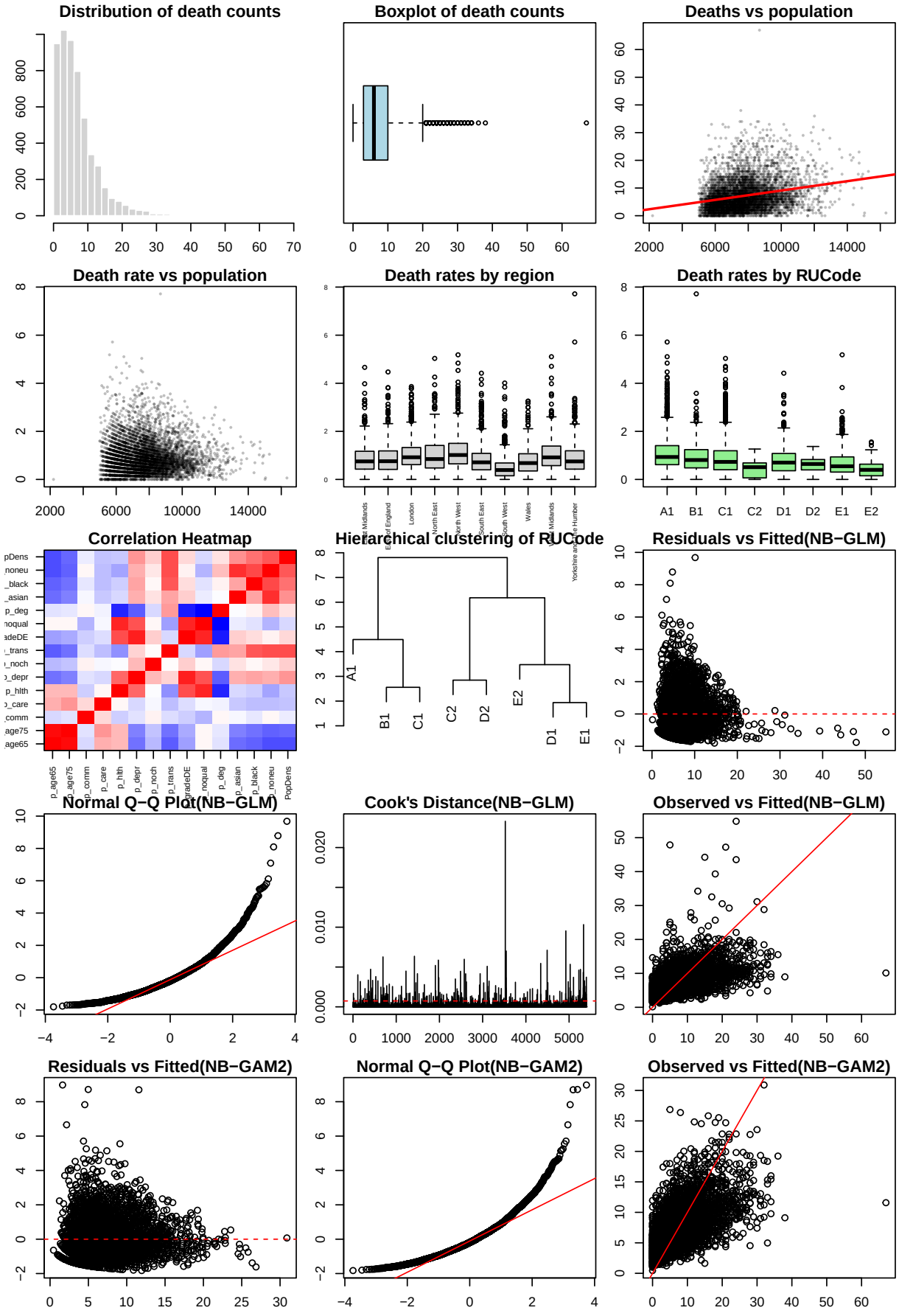


Table 1: Derived variables and their definitions

New_variable	Original_variables	Definition
log_ pop	PopTot	log(PopTot)
p_ age75	age_75_plus, PopTot	age_75_plus / PopTot
p_ comm	PopComm, PopTot	PopComm / PopTot
p_ care	carehome_residents, PopTot	carehome_residents / PopTot
p_ hlth	HH_HealthPrb, HH	HH_HealthPrb / HH
p_ depr	HHDepriv3, HHDepriv4, HH	(HHDepriv3 + HHDepriv4) / HH
p_ noch	HHNoCH, HH	HHNoCH / HH
p_ trans	public_transport_total, PopTot	public_transport_total / PopTot
p_ gradeDE	GradeDE, grade_total	GradeDE / grade_total
p_ noqual	NoQual, qual_total	NoQual / qual_total
p_ deg	Qual4+, qual_total	Qual4+ / qual_total
p_ asian	EthAsian, PopTot	EthAsian / PopTot
p_ black	EthBlack, PopTot	EthBlack / PopTot
p_ noneu	BornNonEU, PopTot	BornNonEU / PopTot
death_ rate_ per_ 1000	Deaths_obs, PopTot	(Deaths_obs / PopTot) * 1000

Contribution

We confirm that all members contributed equally to this project.